

Non-Weak-Versality of the Genus-8 Fano Threefold V_{14} under $\mathrm{PSL}_2(\mathbb{F}_{11})$: The Centralizer Obstruction

Transcribed from the E–Klein-Cubic project derivation record

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Abstract

We give a self-contained account of the “centralizer obstruction,” a theorem proved in the derivation notes of an ongoing investigation into the essential dimension of $\mathrm{PSL}_2(\mathbb{F}_{11})$, and of its application to the second of the two Prokhorov classes of rationally connected $\mathrm{PSL}_2(\mathbb{F}_{11})$ -threefolds, the genus-8 Fano threefold V_{14} . We state and prove, transcribing faithfully from the source, a general theorem: if σ is an involution in a finite group G acting faithfully on a smooth projective variety Y in characteristic 0, if no faithful linear representation of G sends σ to $\pm \mathrm{id}$ (automatic when G is centerless), if $N = C_G(\sigma)$, and if (a) no positive-dimensional irreducible component of Y^σ contains a rational curve, and (b) $Y^N = \emptyset$, then Y is not weakly versal: no G -equivariant rational map from any faithful linear representation of G lands in Y . We then record the machine-verified instantiation of hypotheses (a) and (b) for $Y = V_{14}$, $G = \mathrm{PSL}_2(\mathbb{F}_{11})$, and σ an involution: the centralizer $N = C_G(\sigma)$ is the dihedral group of order 12; V_{14}^σ is a single smooth irreducible genus-1 sextic curve together with two reduced points, hence contains no rational curve; and V_{14}^N is empty. Consequently V_{14} is not weakly versal, its generic twist has no rational point, and V_{14} is not $\mathrm{PSL}_2(\mathbb{F}_{11})$ -unirational — new-theorem territory, since the standard toolbox (fixed-point projection, index-two chords, invariant-hyperplane induction) provably fails on V_{14} , and the Cheltsov–Tschinkel–Zhang equivariant-unirationality program is explicitly scoped to Fano index ≥ 2 , while V_{14} has index 1. Combined with Prokhorov’s theorem that the Klein cubic and V_{14} are the only two equivariant birational classes of rationally connected $\mathrm{PSL}_2(\mathbb{F}_{11})$ -threefolds, and with Duncan–Reichstein’s versality theory for invariant cubic hypersurfaces, this collapses a disjunction: $\mathrm{ed}_{\mathbb{C}}(\mathrm{PSL}_2(\mathbb{F}_{11})) = 3$ if and only if the Klein cubic itself is G -unirational. Whether the Klein cubic is G -unirational remains open. We report exactly what is proved here, what is cited from the literature, and what is machine-verified, and we record precisely where each dependency of the argument stands.

1 Introduction

1.1 Essential dimension and the versality dictionary

Let G be a finite group. A *versal G -variety* is, informally, a G -variety that receives an equivariant rational map from every sufficiently generic G -variety of the same kind; the precise definitions are recalled in Section 2. The *essential dimension* $\mathrm{ed}(G)$ is the minimal dimension of a versal G -variety carrying a faithful (equivalently, generically free, for finite G) G -action [6]. Over the base field $\mathbb{C}$ we write $\mathrm{ed}_{\mathbb{C}}(G)$.

Duncan and Reichstein [6] organize several competing notions of versality — *weakly versal*, *versal*, *very versal* — and prove that they correspond exactly to statements about rational points on twisted forms:

$$\text{stably birationally linear} \implies \text{very versal} \implies \text{versal} \implies \text{weakly versal},$$

with weak versality equivalent to the existence, for every twisting pair (T, K) , of a K -point on the twist ${}^T X$; very versality equivalent to the existence of a linear representation $G \rightarrow GL(V)$ and a G -equivariant *dominant* rational map $V \dashrightarrow X$. In the later terminology of Cheltsov–Tschinkel–Zhang [5], very versality of a projective G -variety X is called *G -unirationality*: existence of a dominant equivariant rational map $\mathbb{P}(V) \dashrightarrow X$ for some G -representation V . The precise statements are recalled in Section 2.

The group $\mathrm{PSL}_2(\mathbb{F}_{11})$ is the automorphism group of the *Klein cubic threefold*

$$Y_{\mathrm{Klein}} = \{x_0^2 x_1 + x_1^2 x_2 + x_2^2 x_3 + x_3^2 x_4 + x_4^2 x_0 = 0\} \subset \mathbb{P}^4,$$

acting via a linear representation $G \rightarrow GL_5(\mathbb{C})$ [1]. Since G is simple, the tautological projection $\mathbb{C}^5 \setminus \{0\} \rightarrow \mathbb{P}^4$ exhibits $\mathbb{P}^4$ (hence Y_{Klein} , once one uses that a smooth cubic hypersurface with a rational point is unirational) as very versal for G , giving $\mathrm{ed}_{\mathbb{C}}(\mathrm{PSL}_2(\mathbb{F}_{11})) \leq 4$; and $\mathrm{PSL}_2(\mathbb{F}_{11})$ cannot act faithfully on a unirational surface, giving $\mathrm{ed}_{\mathbb{C}}(\mathrm{PSL}_2(\mathbb{F}_{11})) \geq 3$ [6, Prop. 10.8]. So the headline question of this circle of ideas is exactly

$$\mathrm{ed}_{\mathbb{C}}(\mathrm{PSL}_2(\mathbb{F}_{11})) \in \{3, 4\},$$

and Duncan–Reichstein already frame it as a disjunction of open conjectures: their Proposition 10.8(b) shows the Cassels–Swinnerton-Dyer conjecture on 0-cycles of degree prime to 3 on cubic hypersurfaces would force $\mathrm{ed}_{\mathbb{C}}(\mathrm{PSL}_2(\mathbb{F}_{11})) = 3$, while part (c) shows Dolgachev’s conjecture $\mathrm{ed} \geq \mathrm{Crdim}$ would force $\mathrm{ed}_{\mathbb{C}}(\mathrm{PSL}_2(\mathbb{F}_{11})) = 4$ (via Prokhorov’s theorem that $\mathrm{PSL}_2(\mathbb{F}_{11})$ admits no faithful action on any rational complex threefold). Settling the headline would also complete Beauville’s classification of finite simple groups of essential dimension 3 [2]. This note bears on neither conjecture directly; its role is to narrow *which variety* the question is really about.

1.2 The two Prokhorov classes, and where V_{14} enters

Prokhorov’s classification of finite simple subgroups of the rank-3 Cremona group contains the following fact, central to everything below.

Fact 1.1 (Prokhorov’s two-class theorem [8, Thm. 1.5]). *There are exactly two equivariant birational equivalence classes of rationally connected complex $\mathrm{PSL}_2(\mathbb{F}_{11})$ -threefolds, represented by the Klein cubic threefold Y_{Klein} and by a smooth Fano threefold V_{14} of Picard rank 1, index 1, and genus 8 (degree 14).*

Consequently, if $\mathrm{ed}_{\mathbb{C}}(\mathrm{PSL}_2(\mathbb{F}_{11})) = 3$, the minimal-dimension versal witness is (after the standard reduction: a versal variety of minimal dimension is very versal, hence unirational, hence rationally connected) equivariantly birational to one of these two classes, and *that* class must itself be G -unirational. The question “is $\mathrm{ed}_{\mathbb{C}}(\mathrm{PSL}_2(\mathbb{F}_{11})) = 3$?” therefore splits into two sub-questions, one about each class. The Klein cubic’s own G -unirationality is untouched by this note and remains open. The present note settles the V_{14} sub-question: V_{14} is *not* G -unirational, and more — not even weakly versal. This removes V_{14} from the disjunction entirely, leaving the Klein cubic as the sole possible witness to $\mathrm{ed}_{\mathbb{C}}(\mathrm{PSL}_2(\mathbb{F}_{11})) = 3$ (Corollary 7.1 below, the collapse of Duncan–Reichstein’s Remark 10.10).

It is worth recording immediately why this needed a new argument. Tschinkel–Zhang [9] show that the classical (non-equivariant) Pfaffian–Grassmannian correspondence between Y_{Klein} and V_{14} upgrades, after stabilizing by $\mathbb{P}^2 \times \mathbb{P}(V)$ for the irreducible 6-dimensional representation V of the double cover $\tilde{G} = \mathrm{SL}_2(\mathbb{F}_{11})$, to a G -equivariant birational equivalence; but the two actions are separately birationally rigid [3, Thm. 4.3], so they are *not* equivariantly birational to each other,

and plain G -unirationality of one does not automatically transfer to the other. The three standard tools that close most comparable cases in the literature — projection from a G -fixed point, chords through the fixed pair of an index-2 subgroup, and induction from a G -invariant hyperplane section whose induced action is G -unirational [5, Prop. 3.5] — all fail structurally on V_{14} : the first because the relevant ambient representation is irreducible (no fixed points), the second because G is simple (no index-2 subgroups), and the third both for the full group (no invariant hyperplane) and, more strongly, for *every* subgroup whatsoever, because V_{14} has Fano index 1 and its hyperplane sections are K3 surfaces, never unirational. Direct computation with several subgroups (the A_5 -fixed locus, the D_{12} -fixed pencil) confirms that the usual fixed-point tools cannot close any V_{14} case either. And Cheltsov–Tschinkel–Zhang’s own equivariant unirationality program [5] is explicitly scoped to Fano threefolds of index ≥ 2 ; V_{14} , at index 1, falls outside it entirely. Nothing in the literature closes the V_{14} case. What follows is a different kind of obstruction — a *necessary* condition for weak versality that requires no dominance, no covariant, and no degree data, checked on a single involution.

1.3 What is proved here

Section 2 recalls the precise definitions (versality, G -unirationality, rationally chain connected varieties, fixed loci) and the three results from the project’s equivariant fixed-locus calculus that the main proof invokes. Section 3 states and proves the centralizer obstruction (Theorem 3.1 below, called Corollary IX.1 in the source project notes, where it appears as a variant of an earlier “central obstruction” with the centralizer of an involution replacing the center of the group). Section 4 records the three inputs needed to instantiate the theorem on V_{14} , all machine-verified. Section 5 summarizes the verification itself: two independent computer-algebra engines, three primes, and an exact characteristic-0 computation. Section 6 draws the conclusion that V_{14} is not weakly versal, with a remark on what this does and does not say about V_{14} as a bare variety. Section 7 proves that this collapses the essential-dimension question onto the Klein cubic alone. Section 8 is a frank accounting of what this note actually establishes and what it depends on. We emphasize at the outset: **the Klein cubic’s own G -unirationality is open**; nothing here proves or suggests a value for $\mathrm{ed}_{\mathbb{C}}(\mathrm{PSL}_2(\mathbb{F}_{11}))$, and the note makes no claim about F_{55} or any other subgroup beyond what is stated.

2 Preliminaries

Throughout, the base field is $\mathbb{C}$ (or any algebraically closed field of characteristic 0), and G is a finite group.

Definition 2.1 (G -variety, equivariant rational map). A G -variety is a smooth projective integral variety X with a regular faithful G -action. For $H \leq G$, $X^H = \{x \in X : hx = x \ \forall h \in H\}$; in characteristic 0, with X smooth, X^H is smooth (by linearizing the H -action at a fixed point), so its irreducible components coincide with its connected components and are pairwise disjoint for fixed H . A rational map $f : X \dashrightarrow Y$ of G -varieties is G -equivariant if it commutes with the two G -actions wherever it is defined; it need not be dominant unless stated.

Definition 2.2 (Versality, following Duncan–Reichstein [6]). Let G be a finite (or, more generally, linear algebraic) group over a field k . A G -action on an irreducible k -variety X is

- *weakly versal* if for every field K/k with K infinite and every G -torsor $T \rightarrow \mathrm{Spec}(K)$, there is a G -equivariant k -morphism $T \rightarrow X$;
- *versal* if every G -invariant dense open subvariety of X is weakly versal;

- *very versal* if there exist a linear representation $G \rightarrow GL(V)$ and a G -equivariant dominant rational map $V \dashrightarrow X$.

Writing (T, K) for a twisting pair and ${}^T X$ for the twist of X_K by T , Duncan–Reichstein’s Theorem 1.1 identifies these notions with statements about rational points: X is weakly versal iff ${}^T X(K) \neq \emptyset$ for every twisting pair (T, K) ; X is versal iff K -points are dense in ${}^T X$ for every twisting pair; X is very versal iff ${}^T X$ is K -unirational for every twisting pair. An equivalent, and often more convenient, characterization of weak versality ([6, Remark 2.1]) is: X is weakly versal if and only if for *every* generically free primitive G -variety Z with $k(Z)^G$ infinite, there is a G -equivariant rational map $Z \dashrightarrow X$. Since every faithful linear representation of a *finite* group is automatically generically free (the union of the proper fixed subspaces V^h , $h \neq 1$, is a proper closed subset, because faithfulness forces each $V^h \subsetneq V$), a faithful linear G -rep V is itself an instance of such a Z ; consequently, if X fails to receive an equivariant rational map from even one faithful linear representation of G , then X is not weakly versal. This is the precise sense in which the source’s informal shorthand “the generic twist of X has no rational point” should be read: by Theorem 1.1(a) it packages the assertion that *some* twisting pair (T, K) produces a twist ${}^T X$ with ${}^T X(K) = \emptyset$.

Definition 2.3 (G -unirationality [5]). A G -variety X is *G -unirational* if there is a dominant G -equivariant rational map $\mathbb{P}(V) \dashrightarrow X$ for some G -representation V . Replacing $\mathbb{P}(V)$ by V shows this notion coincides with very versality in the sense above.

Definition 2.4 (Rationally chain connected). A projective variety Z is *rationally chain connected* (RCC) if two general closed points of Z can be joined by a connected chain of irreducible rational curves on Z . RCC is a birational invariant among proper varieties, and a projective bundle over an RCC base is RCC.

We will also need three results from the project’s equivariant fixed-locus calculus [S2], a working framework built to track how the fixed-locus structure of a G -variety transforms under equivariant blowups. We state them exactly as needed for the proof in Section 3, with pointers to their labels in the source.

Fact 2.5 (Blowup calculus, condensed; [S2, Thm. 2.1]). *Let $Z \subset X$ be a smooth G -stable center of a blowup $\pi : X' = \text{Bl}_Z X \rightarrow X$ with exceptional divisor $E = \mathbb{P}(N_{Z/X})$. Fix $H \leq G$ and an irreducible component F of X^H .*

- (i) *If $F \not\subseteq Z$: the strict transform $F' \subset X'$ (the blowup of F along $F \cap Z$) is again an irreducible component of $(X')^H$, and is equivariantly birational to F .*
- (ii) *If $F \subseteq Z$: H acts on the normal bundle $N_{Z/X}|_F$ covering the identity on F ; write $N_{Z/X}|_F = \bigoplus_\chi N^\chi$ for the decomposition into H -eigen-subbundles (of rank locally constant along F , hence constant since F is irreducible). For each nontrivial character χ , the closure $\mathbb{P}(N^\chi) \subseteq E$ is an irreducible component of $(X')^H$, a projective subbundle of E over F .*

Lemma 2.6 (Rational-chain going-down; [S2, Lem. 4.2]). *Let $q : X \rightarrow Y$ be a proper G -morphism and $F \subseteq X$ an irreducible closed RCC subvariety fixed pointwise by some $H \leq G$. Then $q(F)$ is an irreducible closed RCC subvariety of Y^H . In particular, if the irreducible component of Y^H containing $q(F)$ contains no rational curve (e.g. is a point, or a curve of geometric genus ≥ 1), then $q(F)$ is a single point of it.*

Proof. Chains of rational curves push forward under q to chains of rational curves or points, so a connecting chain between general points of F maps to a connecting chain between general points of

$q(F)$; this shows $q(F)$ is RCC. Pointwise H -fixedness of F gives $q(F) \subseteq Y^H$ by equivariance. If the component of Y^H containing $q(F)$ contains no rational curve, an RCC subvariety of it cannot have positive dimension (a positive-dimensional RCC variety is covered by rational curves), so $q(F)$ is a point. $\square$

Lemma 2.7 (RCC propagation through the calculus; [S2, Lem. 4.3]). *In the setting of Fact 2.5, every exceptional stratum $\mathbb{P}(N^\times)$ is a projectivized bundle over F (there written F_Z), hence RCC whenever F is; and strict transforms preserve RCC-ness (being equivariantly birational to F).*

Correction 2.8 (Scope of Lemma 2.7, recorded in the source as “Correction I-C,” 2026-08-05). Lemma 2.7 is stated above exactly as it should be used: *a single* exceptional stratum, or strict transform, over *a single already-RCC* F , remains RCC after one blowup. The source project’s notes record that a stronger global consequence once drawn from it — that *every* stratum of the fixed-locus decomposition of *every* G -model of X is RCC, as soon as this holds on one model — is **false** for arbitrary towers of equivariant blowups. Two counterexamples are recorded: one due to A. Duncan (blowing up an isolated fixed point with weights $(-1, -1, -1)$, then a smooth plane quartic curve inside the resulting exceptional $\mathbb{P}^2$, produces a genus-3 fixed component on that model); and one found in-house (blowing up the G -orbit of a generic quartic curve inside a fixed hyperplane of $\mathbb{P}(W)$ itself produces, over that curve, an exceptional locus whose fixed part has a section isomorphic to the quartic, again genus 3). The per-blowup statement of Lemma 2.7 is unaffected by these examples; what fails is only the inductive closure of that statement across *arbitrary* models. The induction does close — and the stronger global statement is recovered — when the blowup centers used are (unions of components of) strata of the running model, so-called *stabilizer-stratified towers*. We record this correction because it is exactly the kind of scope restriction that must be checked against any argument citing Lemma 2.7 across a multi-step induction; see Remark 3.3 below for how the proof of Theorem 3.1 fares against it.

Fact 2.9 (Equivariant resolution / weak factorization; [S2, Prop. 3.3]). *In characteristic 0, any G -equivariant proper birational morphism (in particular, the resolution of the graph of a G -equivariant rational map $X \dashrightarrow Y$ of G -varieties) can be dominated by a tower of blowups of smooth projective G -varieties along smooth G -invariant centers, resolving it to a morphism. This rests on Hironaka’s resolution of singularities made canonical (hence automatically equivariant) in characteristic 0, together with the equivariant form of the Abramovich–Karu–Matsuki–Włodarczyk weak factorization theorem, which the source cites under the abbreviation “AKMW.” Concretely: if $f : X \dashrightarrow Y$ is G -equivariant and $\Gamma \subseteq X \times Y$ is the closure of its graph with the diagonal G -action, an equivariant resolution $\tilde{\Gamma} \rightarrow \Gamma$ has first projection $\tilde{\pi} : \tilde{\Gamma} \rightarrow X$ a tower of blowups along smooth G -invariant centers, and second projection $\tilde{f} : \tilde{\Gamma} \rightarrow Y$ an honest (everywhere defined) G -morphism extending f .*

Finally, we fix notation for fixed loci used throughout: for $H \leq G$ and a G -variety Y , Y^H denotes the (smooth, by the definition above) fixed locus; $Y^\sigma := Y^{(\sigma)}$ for an element σ ; and for a subgroup $N \leq G$, Y^N is the locus fixed pointwise by all of N . Since $N \leq G$ implies $Y^G \subseteq Y^N$ (anything fixed by all of G is in particular fixed by all of N), $Y^N = \emptyset$ is a strictly stronger statement than $Y^G = \emptyset$.

3 The centralizer obstruction

The theorem below is Corollary IX.1 of the source project notes. It is recorded there under that name because it descends from an earlier result in the same project, the “central obstruction” (Corollary T3.1): the special case where $\sigma = z$ is taken to be a nontrivial *central* element of G ,

where $N = C_G(z) = G$ automatically, and where hypothesis (b) reads $Y^G = \emptyset$. Corollary T3.1 was designed for groups with nontrivial center; Theorem 3.1 below is the variant anticipated for *local* stabilizers, replacing the center by the centralizer of a single involution. The trade-off is that the eigenspace splitting of a representation is only stable under $N = C_G(\sigma)$ rather than under all of G , so the conclusion is correspondingly narrower: it applies specifically to *linear* sources (where a single global N -stable eigenspace is available before any resolution is performed), rather than to arbitrary rationally connected sources carrying a G -stable fixed stratum on some model.

Theorem 3.1 (Centralizer obstruction). *Let G be a finite group acting faithfully on a smooth projective variety Y over an algebraically closed field of characteristic 0, and let $\sigma \in G$ be an involution such that no faithful linear representation $\rho : G \rightarrow GL(V)$ under consideration degenerates to $\rho(\sigma) = \pm \text{id}$. (This is automatic whenever G is centerless: if $\rho(\sigma) = \text{id}$ then $\sigma \in \ker \rho = \{1\}$ by faithfulness, contradicting that σ is an involution; and if $\rho(\sigma) = -\text{id}$, then since $-\text{id}$ is central in $GL(V)$, for every $g \in G$ we get $\rho(g\sigma g^{-1}) = \rho(g)(-\text{id})\rho(g)^{-1} = -\text{id} = \rho(\sigma)$, so $g\sigma g^{-1} = \sigma$ by faithfulness, i.e. $\sigma \in Z(G)$; if $Z(G) = 1$ this again forces $\sigma = 1$, a contradiction.) Set $N = C_G(\sigma)$. Assume:*

- (a) *no positive-dimensional irreducible component of Y^σ contains a rational curve;*
- (b) $Y^N = \emptyset$.

Then for every faithful linear representation V of G , there is no G -equivariant rational map $\mathbb{P}(V) \dashrightarrow Y$ — dominant or not — and none $V \dashrightarrow Y$ either (the second follows from the first by embedding $V \subset \mathbb{P}(V \oplus \text{triv})$, again linear and faithful). In particular Y is not weakly versal: some twist of Y has no rational point.

Proof. Fix a faithful linear representation $\rho : G \rightarrow GL(V)$. Since σ is an involution, $\rho(\sigma)^2 = \text{id}$, so $\rho(\sigma)$ is diagonalizable with eigenvalues in $\{+1, -1\}$; by hypothesis it is neither $+\text{id}$ nor $-\text{id}$, so both eigenspaces V_+ and V_- are nonzero, giving $V = V_+ \oplus V_-$. Consider $\mathbb{P}(V_+) \subset \mathbb{P}(V)$: it is nonempty (as $V_+ \neq 0$), irreducible and rational (a linear subspace of projective space), pointwise fixed by σ (by definition of the $+1$ -eigenspace, so certainly RCC), and N -stable: for $n \in N$ and $v \in V_+$, since n commutes with σ in G we have $\sigma \cdot (n \cdot v) = n \cdot (\sigma \cdot v) = n \cdot v$, so $n \cdot v \in V_+$ too. This is exactly the point at which the *linearity* of the representation is used: it supplies a single subspace V_+ , stable under N , valid at every point of $\mathbb{P}(V)$ simultaneously, rather than a subvariety that would have to be located after the fact.

Suppose, for contradiction, that $\varphi : \mathbb{P}(V) \dashrightarrow Y$ is a G -equivariant rational map (we do not assume it is dominant, or even non-constant). By Fact 2.9, resolve the graph of φ equivariantly: there is a tower of blowups $\pi : \tilde{X} \rightarrow \mathbb{P}(V)$ along smooth G -invariant centers, together with an honest G -morphism $\tilde{\varphi} : \tilde{X} \rightarrow Y$ extending φ .

Claim. $\tilde{X}$ carries an irreducible, N -stable, pointwise- σ -fixed, RCC closed subvariety $\tilde{F}$.

Proof of Claim, by induction up the tower. Set $F_0 = \mathbb{P}(V_+) \subseteq \mathbb{P}(V)$, the base of the tower; by the paragraph above it is irreducible, RCC (being rational), N -stable, and pointwise σ -fixed. Suppose F has been constructed with these four properties at some intermediate stage X_i of the tower, and let $Z \subset X_i$ be the (smooth, G -invariant, hence also σ -invariant and N -invariant) center of the next blowup $X_{i+1} = \text{Bl}_Z X_i \rightarrow X_i$, with exceptional divisor $E = \mathbb{P}(N_{Z/X_i})$.

Case $F \not\subset Z$. Let $F' \subset X_{i+1}$ be the strict transform of F , i.e. the blowup of F along $F \cap Z$ (equivalently, the closure of the preimage of $F \setminus Z$). By Fact 2.5(i), F' is irreducible and equivariantly birational to F , hence RCC by Lemma 2.7. It is N -stable, since N preserves both F (inductive hypothesis) and Z (G -invariance), hence preserves $F \cap Z$ and the blowup construction. It is pointwise

σ -fixed: the open dense subset $F \setminus Z$ of F' (via the isomorphism $F \setminus Z \xrightarrow{\sim} F' \setminus E$) is pointwise σ -fixed by the inductive hypothesis, and being pointwise fixed by a fixed automorphism is a closed condition, so it persists on the closure F' .

Case $F \subseteq Z$. Since σ fixes F pointwise and preserves Z , it acts on the total space of the normal bundle $N_{Z/X_i}|_F$ covering the identity map on F ; being an involution in characteristic 0, this action splits the bundle into eigen-subbundles $N_{Z/X_i}|_F = N_+ \oplus N_-$ (ranks constant along the irreducible base F). As Z has positive codimension in X_i , at least one of N_+, N_- is nonzero; choose $\lambda \in \{+, -\}$ with $N_\lambda \neq 0$ (if both are nonzero, take $\lambda = +$, fixing a convention), and set

$$F' = \mathbb{P}(N_\lambda|_F) \subset E|_F \subset X_{i+1}.$$

This is a projective subbundle of E over F (Fact 2.5 (ii), applied to the eigen-character λ), hence irreducible, and RCC by Lemma 2.7 (a projective bundle over the RCC base F). It is pointwise σ -fixed: a point of $\mathbb{P}(N_\lambda|_F)$ is a line in a fiber of N_λ , and σ scales every vector in that fiber by the fixed scalar $\lambda = \pm 1$, hence fixes the *line* regardless of the sign. It is N -stable: N preserves F and Z , hence acts on $N_{Z/X_i}|_F$ covering its action on F ; since every $n \in N$ commutes with σ , n carries the λ -eigenspace of σ at a point $p \in F$ to the λ -eigenspace at $n \cdot p$ (exactly the same commuting argument as for V_+ above), so n preserves the subbundle N_λ and hence the subvariety $F' = \mathbb{P}(N_\lambda|_F)$.

In both cases F' again has all four properties, completing the induction; after finitely many steps (the tower is finite) we obtain $\tilde{F} \subseteq \tilde{X}$ with all four properties, proving the Claim.

Now apply $\tilde{\varphi}$. Since $\tilde{F}$ is pointwise σ -fixed and $\tilde{\varphi}$ is G -equivariant, every $x \in \tilde{F}$ satisfies $\tilde{\varphi}(x) = \tilde{\varphi}(\sigma x) = \sigma \tilde{\varphi}(x)$, so $\tilde{\varphi}(\tilde{F}) \subseteq Y^\sigma$. Since $\tilde{\varphi}$ is a morphism of projective varieties (proper) and $\tilde{F}$ is irreducible and closed, $\tilde{\varphi}(\tilde{F})$ is irreducible and closed; since $\tilde{F}$ is RCC, Lemma 2.6 shows $\tilde{\varphi}(\tilde{F})$ is an RCC closed subvariety of Y^σ , lying (by irreducibility, and smoothness of Y^σ) inside a single irreducible component of Y^σ . By hypothesis (a), if that component has positive dimension it contains no rational curve, so by Lemma 2.6 $\tilde{\varphi}(\tilde{F})$ is forced to be a single point $y \in Y^\sigma$.

Finally, $\tilde{F}$ is N -stable and $\tilde{\varphi}$ is G -equivariant (in particular N -equivariant), so for every $n \in N$:

$$n \cdot y = n \cdot \tilde{\varphi}(\tilde{F}) = \tilde{\varphi}(n \cdot \tilde{F}) = \tilde{\varphi}(\tilde{F}) = \{y\},$$

using $n \cdot \tilde{F} = \tilde{F}$ as sets. Hence $y \in Y^N$, contradicting hypothesis (b), $Y^N = \emptyset$. This contradiction shows no such φ can exist, for any faithful linear V , proving the theorem. $\square$

Remark 3.2 (What is not used). The proof never assumes φ is dominant; a would-be *constant* equivariant map is excluded too, by the same argument (its image would be a single G -fixed point, and $Y^G \subseteq Y^N = \emptyset$). No covariant or degree data of any kind enters. The source remarks that this is the “all-degree, search-free” obstruction shape its broader investigation had been pursuing since an earlier stage of the project (there labeled “E34”): a single finite verification, on one involution, ruling out every faithful linear source of every dimension and every degree at once.

Remark 3.3 (The proof and Correction 2.8). It is worth checking explicitly that the induction above survives Correction 2.8, since it invokes Lemma 2.7 inside a multi-step induction along a tower that is an arbitrary equivariant resolution of a graph (not manifestly a stabilizer- stratified tower in the sense of the correction). On inspection it does: at each step the induction tracks a *single* stratum F , and in the case $F \subseteq Z$ it applies Lemma 2.7 with the base of the projective bundle equal to F itself, whose RCC-ness was already established at the previous stage. This is precisely the per-blowup statement that Correction 2.8 leaves intact; the proof never needs the stronger (and, in general, false) global claim that *every* fixed-locus stratum of *every* model is RCC. We record this because it is exactly the sort of check a citation of a scope-corrected lemma demands; the source project notes do not make this observation explicitly at the point where Theorem 3.1 is stated, so we flag it here as an editorial addition rather than a transcribed claim.

4 The V_{14} instantiation

To apply Theorem 3.1 to $Y = V_{14}$ and $G = \mathrm{PSL}_2(\mathbb{F}_{11})$, three facts must be verified: the identification of $N = C_G(\sigma)$, hypothesis (a), and hypothesis (b). Since G is simple (in particular centerless), the degeneracy hypothesis on $\rho(\sigma)$ in Theorem 3.1 holds automatically for *every* faithful linear representation of G , so verifying (a) and (b) once, for V_{14} itself, rules out every linear source simultaneously.

4.1 The ambient realization of V_{14}

The computations below realize V_{14} concretely as a linear section of a Grassmannian, following the classical Pfaffian–Grassmannian correspondence (recalled in Section 6). Let $\tilde{G} = \mathrm{SL}_2(\mathbb{F}_{11})$ be the double cover of G , and let U be its irreducible 6-dimensional (“even Weil”) representation. The central element $-1 \in \tilde{G}$ acts as $-\mathrm{id}$ on U , so U does *not* descend to a linear representation of G itself (only $\mathbb{P}(U)$ carries a genuine, projectively linear, G -action; this is the “spin” source discussed and explicitly set aside in Remark 6.3 below). Its exterior square $\Lambda^2 U$ is 15-dimensional, and since -1 acts on $\Lambda^2 U$ as $(-1)^2 = +1$, *this* representation is center-trivial and does descend to a genuine linear representation of $G = \mathrm{PSL}_2(\mathbb{F}_{11})$. It decomposes as

$$\Lambda^2 U \cong W_5 \oplus M,$$

with W_5 a 5-dimensional and M a 10-dimensional (denoted “10’” in the source, to distinguish it from the other 10-dimensional irreducible of G) summand. The variety

$$V_{14} := \mathrm{Gr}(2, U) \cap \mathbb{P}(M) \subset \mathbb{P}(M) = \mathbb{P}^9$$

is then the G -invariant linear section of the Grassmannian $\mathrm{Gr}(2, 6) \subset \mathbb{P}(\Lambda^2 U) = \mathbb{P}^{14}$ cut by the linear subspace $\mathbb{P}(M)$; its expected dimension is $8 - 5 = 3$ (the Grassmannian has dimension 8, and $\mathbb{P}(M)$ has codimension 5 in $\mathbb{P}^{14}$), and its degree, once smoothness holds, is 14 by transversality of a linear section of the (arithmetically Cohen–Macaulay) Grassmannian.

4.2 Input 1: $N = C_G(\sigma) = D_{12}$

The class of involutions in $G = \mathrm{PSL}_2(\mathbb{F}_{11})$ is a single conjugacy class of size 55. Fixing a representative σ , the centralizer $N = C_G(\sigma)$ has order 12, and its structure — the dihedral group of order 12 (the symmetry group of a hexagon; equivalently, 7 of its 11 nonidentity elements are involutions: 6 “reflections” and one further involution in the order-6 cyclic subgroup) — is a fact about the abstract group G alone, independent of any variety it acts on.

4.3 Input 2: hypothesis (a)

Since $V_{14} \subset \mathbb{P}(M)$ is σ -stable (as $\sigma \in G$ acts linearly on M , preserving V_{14} setwise), and $M = M_+ \oplus M_-$ splits into σ -eigenspaces of dimensions 6 and 4 respectively,

$$V_{14}^\sigma = (V_{14} \cap \mathbb{P}(M_+)) \sqcup (V_{14} \cap \mathbb{P}(M_-)).$$

The sealed computation (Section 5) finds:

- $V_{14} \cap \mathbb{P}(M_+)$ is a curve of degree 6 with Hilbert polynomial $6i$ (so arithmetic genus $p_a = 1$), whose saturated ideal is radical with a single associated prime (i.e. the curve is reduced and irreducible) and whose Jacobian minors saturate to the irrelevant ideal, i.e. the curve is *smooth*. A smooth irreducible curve of arithmetic genus 1 has geometric genus 1 and is therefore not rational.

- $V_{14} \cap \mathbb{P}(M_-)$ is a reduced, 0-dimensional scheme of degree 2 — two points.

So V_{14}^σ consists of one smooth irreducible genus-1 sextic curve together with two reduced points, and **contains no rational curve**: hypothesis (a) holds.

Remark 4.1 (Why smoothness is load-bearing). The Hilbert polynomial $6i$ only certifies the *arithmetic* genus $p_a = 1$ of the sextic; it says nothing about whether the curve is smooth. An arithmetic-genus-1 curve with a single node has geometric genus 0 — its normalization is $\mathbb{P}^1$ — and is therefore *rational*. Had the sextic in $V_{14} \cap \mathbb{P}(M_+)$ turned out to be such a nodal curve, hypothesis (a) would fail outright (a rational curve would sit inside a positive-dimensional component of V_{14}^σ), and the entire argument of this section would collapse. This is exactly the trap the source project notes flag explicitly when setting up the verification: dimension, degree, and Hilbert polynomial alone do not suffice; the smoothness of the sextic must be checked directly (via the Jacobian criterion), and it was.

4.4 Input 3: hypothesis (b)

The subgroup $N = D_{12}$ acts on M (10-dimensional); a point of $\mathbb{P}(M)$ is fixed pointwise by all of N exactly when it spans a line on which N acts by some (possibly nontrivial) character, so V_{14}^N decomposes across the isotypic pieces of $M|_N$ for the four 1-dimensional characters of D_{12} (the two-dimensional irreducibles of the nonabelian group D_{12} contain no N -stable lines, hence contribute nothing). These four pieces have dimensions 2, 1, 1, 0 (the dimension-2 piece is the space of N -invariants proper, i.e. the trivial-character piece, matching the pencil found in earlier, exploratory computation on the same group action). The sealed computation finds $V_{14} \cap \mathbb{P}(\text{piece}) = \emptyset$ in *all four* pieces (the fourth, being already 0-dimensional, is vacuously empty). Equivalently: restricted to the pencil spanned by the trivial-character piece, the relevant Plücker data has rank 6 at all but three points of the pencil, and rank 4 (never rank 2, which is what membership in $\text{Gr}(2, 6)$ would additionally require there) at those three special points. Hence

$$V_{14}^{D_{12}} = \emptyset,$$

and hypothesis (b) holds.

5 Machine verification

The three inputs above were sealed by a director-run verification (project packet [goal_runs_after_c53d89a/FIX_IX_SEAL](#), 2026-08-06), using two independent computational engines — exact rational/modular linear algebra in Python, and Macaulay2 for the ideal-theoretic computations (saturation, primary decomposition markers, Jacobian criteria, Hilbert polynomials) — run at two primes, 397 and 199, with an independently-written verifier replaying the entire computation end-to-end at a third, fresh prime, 353, and finally an exact characteristic-0 computation over the cyclotomic field $K = \mathbb{Q}(\zeta_{11})$ (entries of the explicit matrices lie in $\mathbb{Z}[\zeta_{11}, 1/11]$).

5.1 The group model

The representation U is constructed explicitly (an “even-function” model on $\mathbb{F}_{11}$ with a normalized Fourier/Weil generator) via two 6×6 matrix generators, called T_6, S_6 in the packet. At every prime and in the exact field K , the following were checked and passed:

- $\langle T_6, S_6 \rangle$ closes to a group of order exactly $1320 = 2 \cdot 660$, i.e. (the double cover) $\text{SL}_2(\mathbb{F}_{11})$;

- the Weil normalization $S_6^2 = -I$ and (writing gauss for the quadratic Gauss sum entering the construction) $\text{gauss}^2 = -11$;
- the *projective* order profile of the 1320 elements of the model — i.e. the multiset of orders of their images in $G = \text{PSL}_2(\mathbb{F}_{11})$ — is

$$(\text{order } 1 : 2), (2 : 110), (3 : 220), (5 : 528), (6 : 220), (11 : 240),$$

summing to 1320 as required; this matches G 's known element-order spectrum $\{1, 2, 3, 5, 6, 11\}$.

The isotypic projector onto the 10-dimensional summand M of $\Lambda^2 U$ has rank exactly 10 and is idempotent on its own column space; $\text{Ann}(M) \subset \Lambda^4 U$ (under the natural pairing $\Lambda^2 U \times \Lambda^4 U \rightarrow \Lambda^6 U \cong k$) has dimension exactly 5. The identification $\Lambda^2 U \cong W_5 \oplus M$, with M the “10” summand rather than the other 10-dimensional irreducible of G , was confirmed by *two independent methods*: character (isotypic- projector) averaging, and, in the fresh-prime verifier, an independent trace-sum method that compares the sum of $\text{tr}(\Lambda^2 g)$ over each projective-order class of the 1320-element model against the banked character table of $\text{PSL}_2(\mathbb{F}_{11})$ for all four candidate decompositions (W_5 or its Galois twin, paired with either 10-dimensional irreducible); only the pairing with the “10” candidate matches at every class.

5.2 The fixed loci and the ambient variety

At each prime and in K : the σ -eigenspace splitting of M has dimensions $(6, 4)$; the sextic curve $V_{14} \cap \mathbb{P}(M_+)$ has Hilbert polynomial $6i$, is radical with one associated prime, and is smooth (Jacobian minors saturate to the irrelevant ideal); $V_{14} \cap \mathbb{P}(M_-)$ is a reduced degree-2 zero-dimensional scheme. (As a small but genuine cross-check of independence between primes: the two points of $V_{14} \cap \mathbb{P}(M_-)$ are individually rational over $\mathbb{F}_{397}$ and $\mathbb{F}_{353}$, but only Galois-conjugate over $\mathbb{F}_{199}$ — the scheme is degree-2 reduced at every prime regardless, which is all hypothesis (a) requires.) The centralizer $N = C_G(\sigma)$ has order 12 with exactly 7 projective involutions, matching D_{12} ; the four linear-character pieces of $M|_N$ have dimensions $(2, 1, 1, 0)$, and V_{14} meets none of them.

The ambient variety itself was independently confirmed smooth of pure dimension 3: a dual system J , consisting of the Pfaffian sextic (“Pf₆”) together with an adjoint-bivector condition on $\text{Ann}(M)$, is empty at every prime and over K . Since any singular or excess-dimension point of $\text{Gr}(2, U) \cap \mathbb{P}(M)$ would produce a point of J , emptiness of J proves V_{14} is smooth of pure dimension 3; degree 14 then follows by transversality, and irreducibility from the Enriques–Severi–Zariski principle for linear sections of the (projectively normal) Grassmannian. A direct Gröbner-basis computation of $\text{Gr}(2, U) \cap \mathbb{P}(M)$ cross-checks dimension 3 and degree 14 directly, at all three primes. Finally, the cubic Pf₆ on $\mathbb{P}(\text{Ann } M) \cong \mathbb{P}^4$ is G -invariant and nonzero at every prime; by the (separately sealed) uniqueness of the G -invariant cubic on the 5-dimensional representation W_5 , the hypersurface $\{\text{Pf}_6 = 0\}$ is the Klein cubic threefold. This is the Pfaffian-partner identification underlying the classical correspondence invoked in Section 6.

5.3 Status of the seal

The packet’s own exit marker is `FIX-IX-SEAL-PASS`. All checks above appear, with explicit `PASS` markers, in the packet’s log files (`results/checks_397.log`, `checks_199.log`, `checks_353.log`, `checks_K.log`, and the per-prime `m2_sigma_*.out`, `m2_d12_*.out`, `m2_smooth_*.out`, `m2_ambient_*.out` Macaulay2 transcripts), and the independent end-to-end verifier at the fresh prime 353 reports `ALLGREEN`.

6 Corollary: V_{14} is not weakly versal

Corollary 6.1. *The action of $G = \mathrm{PSL}_2(\mathbb{F}_{11})$ on V_{14} is not weakly versal: no G -equivariant rational map from any faithful linear representation of G lands in V_{14} , and (by the Duncan–Reichstein dictionary, Theorem 1.1(a) of [6]) some twist of V_{14} has no rational point. In particular V_{14} is not G -unirational.*

Proof. $G = \mathrm{PSL}_2(\mathbb{F}_{11})$ is simple, hence centerless, so the degeneracy hypothesis of Theorem 3.1 holds automatically for every faithful linear representation of G (Section 4). Section 4 verifies $N = C_G(\sigma) = D_{12}$ and hypotheses (a) and (b) of Theorem 3.1 for $Y = V_{14}$, at two primes, a fresh verifying prime, and in exact characteristic 0 (Section 5). Theorem 3.1 applies directly. $\square$

This is new-theorem territory: as recalled in the introduction, Cheltsov–Tschinkel–Zhang’s equivariant unirationality program is scoped to Fano threefolds of index ≥ 2 , and V_{14} has index 1, so no result in that literature addresses the V_{14} action at all; and the standard fixed-point toolbox fails on V_{14} structurally, as recalled in the introduction.

Remark 6.2 (The obstruction is purely equivariant). It is important to be precise about what Corollary 6.1 does, and does not, say about V_{14} as a bare variety, forgetting the G -action. By the classical Pfaffian–Grassmannian correspondence recalled in Section 5 (of which the G -equivariant identification $\{\mathrm{Pf}_6 = 0\} = Y_{\mathrm{Klein}}$ is the G -invariant instance), V_{14} and Y_{Klein} are *birational to each other as bare varieties* — Tschinkel–Zhang state this explicitly as “well-known classically” [9], and it is the non-equivariant shadow of their Theorem 1.1. Since birational invariants transfer across this equivalence: V_{14} is *unirational* as a bare variety (Kollár’s theorem [7], that a smooth cubic hypersurface of dimension ≥ 2 with a rational point is unirational (recalled as [6, Thm. 10.3]), applied to Y_{Klein} and transported across the birational equivalence), in particular rationally connected — consistent with Fact 1.1 quantifying over “rationally connected $\mathrm{PSL}_2(\mathbb{F}_{11})$ -threefolds.” But V_{14} is *not rational*: Y_{Klein} , being a smooth cubic threefold, is never rational, by the Clemens–Griffiths intermediate-Jacobian obstruction [4], a theorem with no exceptions among smooth cubic threefolds; rationality is a birational invariant, so V_{14} is not rational either. This is exactly Duncan–Reichstein’s own summary of the pair: “they are unirational . . . but not rational” [6, Remark 10.10].

So Corollary 6.1 is emphatically *not* a repackaging of the Clemens–Griffiths obstruction, and does not assert that V_{14} is rational — it is not. What Theorem 3.1 adds is strictly finer, and genuinely equivariant: even restricting attention to sources $\mathbb{P}(V)$ for faithful linear representations V of G — sources that are themselves rational, so that the bare-variety obstruction to rationality is simply not in play on the source side either — no G -equivariant map to V_{14} exists. Weak versality is a strictly G -equivariant refinement of unirationality: a variety can be perfectly unirational as a bare variety (as V_{14} is) while its G -action fails to be weakly versal, exactly as Theorem 3.1 shows here. Nothing in this note bears on the (settled, negative) question of whether V_{14} is rational as a bare variety; the content is entirely about the nonexistence of *equivariant* maps.

Remark 6.3 (The spin flank is not addressed here). Theorem 3.1 quantifies only over *linear* representations of G itself. The 6-dimensional U used to build the ambient space in Section 4 is not such a representation: $-1 \in \tilde{G} = \mathrm{SL}_2(\mathbb{F}_{11})$ acts on U as $-\mathrm{id}$, so only $\mathbb{P}(U)$ carries a (projectively linear) G -action, and U itself is not a source Theorem 3.1 rules out. The source project notes examine this “spin flank” separately and find that, at the level of a single involution σ , it is *not* obstructed by the same argument: the lift $\tilde{\sigma} \in \tilde{G}$ has order 4, and the D_{12} -stable pair of isolated points in V_{14}^σ found in Section 4 matches exactly the shape the spin argument would require rather than forbid. This does not affect Corollary 6.1 or Corollary 7.1 below, both of which quantify only over honest linear representations of G ; but it does mean that this note says nothing about whether $\mathbb{P}(U)$, or other

projectively-linear “spin” sources, admit equivariant maps to V_{14} . That question is left open by the source project notes as well.

7 Corollary IX.2: the disjunction collapses

Corollary 7.1 (Corollary IX.2 of the source). $\text{ed}_{\mathbb{C}}(\text{PSL}_2(\mathbb{F}_{11})) = 3$ if and only if the Klein cubic threefold Y_{Klein} is G -unirational.

This corollary’s inputs are, explicitly: Prokhorov’s two-class theorem (Fact 1.1); Duncan–Reichstein’s Theorem 10.5, that for a finite group G acting on a smooth G -invariant cubic hypersurface $X \subset \mathbb{P}(V)$ with $\dim V \geq 4$, the three properties very versal, versal, and weakly versal are *equivalent* (unconditionally, with no conjectural input needed for just this equivalence); and Corollary 6.1 of this note.

Proof. ($\Rightarrow$) Suppose $\text{ed}_{\mathbb{C}}(\text{PSL}_2(\mathbb{F}_{11})) = 3$. Then there is a 3-dimensional versal G -variety; the standard witness realizing the minimal dimension (a suitable compression of a generically free linear representation) is unirational, hence rationally connected ([6, Remark 2.6]). By Fact 1.1, a rationally connected $\text{PSL}_2(\mathbb{F}_{11})$ -threefold is G -birational to either Y_{Klein} or V_{14} ; since versality is a G -birational invariant, one of these two threefolds is itself versal. It cannot be V_{14} : a versal variety is in particular weakly versal, and Corollary 6.1 shows V_{14} is not weakly versal. So Y_{Klein} is versal, hence (Duncan–Reichstein Theorem 10.5, since $Y_{\text{Klein}} \subset \mathbb{P}^4 = \mathbb{P}(W_5)$ with $\dim W_5 = 5 \geq 4$ is a smooth G -invariant cubic hypersurface) very versal, i.e. G -unirational.

($\Leftarrow$) The source records the converse direction as established separately in the project notes (there labeled “E37”); we do not reproduce it here, in keeping with this note’s brief to transcribe Note IX faithfully rather than to re-derive. We note only, for orientation, that Duncan–Reichstein’s own Proposition 10.8 already records the unconditional bound $3 \leq \text{ed}_{\mathbb{C}}(\text{PSL}_2(\mathbb{F}_{11})) \leq 4$ (Section 1.1 above), which is visibly compatible with — though does not by itself prove — this direction. $\square$

As the source puts it, the essential-dimension question becomes *single-target*: Dolgachev’s conjectural route to $\text{ed}_{\mathbb{C}}(\text{PSL}_2(\mathbb{F}_{11})) = 4$ is exactly the assertion “Klein cubic negative,” and the Cassels–Swinnerton-Dyer route to $\text{ed}_{\mathbb{C}}(\text{PSL}_2(\mathbb{F}_{11})) = 3$ is exactly “Klein cubic positive”; V_{14} can no longer independently supply an $\text{ed}_{\mathbb{C}} = 3$ witness. **We emphasize again: whether Y_{Klein} is G -unirational is open.** This note proves nothing about it, in either direction.

8 Status and dependencies

We record here, as plainly as possible, what this note establishes and what it rests on, distinguishing throughout between what is proved in this note (by transcription from the source), what is cited from published literature, and what is machine-verified.

- **Proved here (transcribed).** Theorem 3.1 (the centralizer obstruction) and its proof; Corollary 6.1 and Corollary 7.1 and their proofs (with the converse direction of Corollary 7.1 explicitly not reproduced, see above). These are faithful transcriptions of Corollary IX.1 and Corollary IX.2 of the source project note `theory/FIX_IX_v14.md`, itself marked “DRAFT-FOR-DERIVATION” at the top of that document.
- **Depends on a project-internal lemma layer, not published literature.** The proof of Theorem 3.1 rests on Fact 2.9 (equivariant resolution/weak factorization), Lemma 2.6 (rational-chain going-down), and Lemma 2.7 (RCC propagation, together with its scope

correction, Correction 2.8), all from the same project’s `theory/FIX_I_bcomplex.md`. That note is itself explicitly marked, at its own top: “Status: DRAFT-FOR-DERIVATION — every statement here is to be treated as claimed-until-derived-or-checked,” with an internal acceptance-test suite (its Section 6, tests T1–T5) serving as its own validation gate. This is derivation-grade project work, not peer-reviewed or published mathematics; Fact 2.9 rests in turn on genuinely standard results (Hironaka resolution made canonical in characteristic 0, and the Abramovich–Karu–Matsuki–Włodarczyk equivariant weak factorization theorem), but the specific formulations of Lemma 2.6 and Lemma 2.7, and their packaging into Theorem 3.1, have not been externally reviewed. We verified directly (Remark 3.3) that the one place Lemma 2.7 is used inside a nontrivial induction survives its own recorded scope correction; we did not independently re-derive Fact 2.9, Lemma 2.6, or the unmarked parts of Lemma 2.7 from first principles.

- **Machine-verified.** The geometric inputs of Section 4 — $N = C_G(\sigma) = D_{12}$, hypothesis (a), and hypothesis (b) — are verified computationally, as detailed in Section 5: two independent engines, two primes (397, 199) plus an independently-coded fresh-prime verifier (353), and an exact characteristic-0 computation over $K = \mathbb{Q}(\zeta_{11})$. This is strong evidence by the standards of computer-algebra verification (independent engines, independent prime, exact characteristic 0, and a genuinely independent second method for the $\Lambda^2 U = W_5 \oplus M$ identification), but it is a director-run verification internal to the project, not an independently reproduced or peer-reviewed computation.
- **Cited, not reproved.** Fact 1.1 (Prokhorov’s two-class theorem), Duncan–Reichstein’s Theorem 10.5 and the versality dictionary of Theorem 1.1, Kollár’s unirationality theorem for cubic hypersurfaces, the Clemens–Griffiths non-rationality theorem, and the identification of $G = \mathrm{PSL}_2(\mathbb{F}_{11})$ as the automorphism group of the Klein cubic via a linear representation [1], are all taken from the literature as stated, without independent re-proof. The Pfaffian–Grassmannian correspondence and the equivariant birationality of Tschinkel–Zhang [9] are likewise cited, used here only for the (non-equivariant) fact that V_{14} and Y_{Klein} are birational as bare varieties (Remark 6.2).
- **Consequence, for context.** Corollary 7.1 makes the essential-dimension question “ $\mathrm{ed}_{\mathbb{C}}(\mathrm{PSL}_2(\mathbb{F}_{11})) = 3$ or 4?” single- target on the Klein cubic threefold: it is now exactly the question of whether Y_{Klein} is G -unirational. **This question is open.** This note does not claim, suggest, or make progress toward a value of $\mathrm{ed}_{\mathbb{C}}(\mathrm{PSL}_2(\mathbb{F}_{11}))$; it does not establish $\mathrm{ed}_{\mathbb{C}}(\mathrm{PSL}_2(\mathbb{F}_{11})) = 4$, and it says nothing whatsoever about F_{55} or any other subgroup of G beyond $N = D_{12} = C_G(\sigma)$ and the group G itself. Any claim that the Klein cubic case, or the value of $\mathrm{ed}_{\mathbb{C}}(\mathrm{PSL}_2(\mathbb{F}_{11}))$, is resolved would go beyond what is proved here and beyond what the source project notes claim.

References

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- [S1] *In-repo project source*. `problems/E-klein-cubic/theory/FIX_IX.v14.md`, Section 5 (“The centralizer obstruction — the keystone, and the linear-source case”), the primary source for Theorem 3.1–Corollary 7.1; Sections 1–4 and 6 for surrounding context.
- [S2] *In-repo project source*. `problems/E-klein-cubic/theory/FIX_I.bcomplex.md`, the source for Fact 2.9 (Prop. 3.3), Lemma 2.6 (Lemma 4.2), Lemma 2.7 (Lemma 4.3), and its scope correction (“Correction I-C”).
- [S3] *In-repo project source*. `problems/E-klein-cubic/theory/FIX_T_gate.md`, the source for Corollary T3.1 (the “central obstruction”), the antecedent of Theorem 3.1 discussed at the start of Section 3.
- [S4] *In-repo machine verification packet*. `problems/E-klein-cubic/goal_runs_after_c53d89a/FIX_IX_SEAL/` (REPORT.md, brief.md, results/, scripts/, verifier.py), underlying Section 5.